ENGR 225 Design Project Part 1

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Assigned: 26 Oct 2026

Due: 6 Nov 2026 (11:59pm on Moodle)

Problem Statement

The overall goal of this two-part project is to design the truss system to support a roadway that will span a canyon. To design the road itself, you will utilize realistic constraints based on engineering standards in part 1. This in turn will inform the design and analysis of the truss system to support that roadway in part 2. There will be many assumptions and simplifications to aide in your analysis.

This first part will focus on the roadway itself. The main goal for part 1 is to select the material and overall size of the road that needs to be constructed in order to meet the below criteria. As an initial note and reminder, you are not looking to make the full engineering design that you could take to a construction crew, but instead are focusing on the selection of some key parameters that would govern the design. The actual design of a roadway is undertaken by many engineers and is subject to multiple regulations and requirements. The goal of this project is to understand the level of complexity involved in making engineering decisions, and to utilize real standards in constraining your initial design.

Design Criteria

- The roadway to be supported by the bridge will be 60 meters long.
- Two lanes of traffic will be utilized in each direction.
- There is no need for a sidewalk or pedestrian walkways.
- This will be used for high-traffic conditions. As example, consider this to be comparable to a new section of bridge on highway 2 just outside of Hartford, or an elevated roadway on highway 9.

Design Goals

- Meet the appropriate standards for the design of the roadway.
- Minimize environmental impact of the materials selected.
- Minimize the cost of materials selected.

Things to Address and Assume

- Based on the design criteria, how would the roadway be classified? What sort of volume of traffic/trucks would you estimate?
- Based on the criteria and your classification, what sort of materials are needed to create the roadway?
- What thickness of these materials is required?
- What is the width of the roadway to meet the above requirements?
- What factors can contribute to environmental impacts from the choice of material? Think about the components to create the materials, their manufacturing, etc.

Resources to Utilize

These are sources that can provide some information and context on how roadways are designed.

- The U.S. Department of Transportation: Standards Specifications
- Connecticut Department of Transportation: Highway Design Manual
- The Bureau of Transportation Statistics: Repository and Open Science Access Portal
- CT DOT Pavement Design Handbook: pdf here
- Others as you see fit to address cost, environmental impacts, etc. The above list is not exhaustive, and you may find clearer information in your own search.

Reporting Requirements

Provide a report, no longer than 4 pages, that clearly addresses the following points:

- (25 pts) Provide the overall geometry and materials that comprise the roadway.
- (25 pts) Quantify the environmental impact of the materials for the roadway. There are a multitude of factors that can contribute to this (CO_2 emissions, mining of materials, potential transportation of materials, etc.), so do your best to try and cover as much as possible. Make a comparison to other alternative materials or projects to provide perspective.
- (10 pts) Estimate the cost of the raw materials for the roadway.
- (40 pts) For each of the graded items above, provide a justification based on your reading and research. Link to standards, best practices, or other resources for every decision you made. Be clear in how these resources informed your decisions. Format this in a consistent manner with a bibliography using IEEE reference style.
  - A quick style guide can be found here.
  - The formal documentation on IEEE style is found here.
- Don't forget to submit your coversheet with this.

A Little Additional Help

There is no way to become completely knowledgeable or an expert on this topic in the little time we have, so I'd encourage you to use AI to help with summarizing the requirements and resources. Cite where information comes from, and be clear with how you used AI, but use your critical thinking with the implementation of AI-based tools to get your design together.